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A Three-Tier Knowledge Graph RAG for Vietnamese Legal Question Answering

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Abstract. Accurate legal question answering demands navigating hierarchical document structures, following cross-references, and multi-hop reasoning, capabilities that standard Retrieval-Augmented Generation (RAG) systems lack. Vietnamese legal text compounds this challenge through tonal ambiguity, domain-specific abbreviations, and rigid structural hierarchies. This paper presents a three-tier knowledge graph RAG architecture that integrates a domain ontology, a knowledge graph, and a document tree forest. Three retrieval tiers, LLM-guided tree traversal, dual-level semantic search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion with knowledge graph expansion, operate over these representations. Evaluation on 530 expert-annotated questions across enterprise law, traffic safety law, and postgraduate education regulation shows that the proposed system retrieves at least one correct article in the top 10 results for 92.2% of queries, outperforming the strongest baseline by 7.4%. These results hold across all three domains without domain-specific tuning. To the best of the authors' knowledge, no prior work combines ontology-enhanced knowledge graphs with multi-tier retrieval for Vietnamese legal question answering.

Keywords. Knowledge Graph, Legal Question Answering, Retrieval Augmented Generation, Ontology, Vietnamese NLP, Multi-Tier Retrieval, Multi-Domain

1. Introduction

Legal forums and social networks in Vietnam are flooded with questions on business formation, traffic penalties, and regulatory compliance. Yet answers are often fragmented, contradictory, or cite repealed provisions.

Legal documents exhibit complex hierarchical structures: a single law organizes content into parts, chapters, sections, articles, clauses, and points, where each level carries distinct legal scope. They also contain pervasive cross-references: answering "If a member borrows money to meet charter capital and later withdraws it, what are the penalties?" requires chaining Article 47 (member obligations) to Article 16, Clause 5 (prohibited acts) across different chapters. Vietnamese compounds this challenge through tonal ambiguity, domain-specific abbreviations (*HDQT*, *CTCP*), and a rigid legal hierarchy (Constitution, Codes, Decrees, Circulars).

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Standard RAG [1] retrieves by vector similarity [2], an approach effective in many domains [3] but inadequate for legal text: embeddings blur precise legal meanings (semantic gap), flat retrieval ignores document hierarchy (no structural awareness), and dense search cannot chain cross-references (no multi-hop reasoning). LightRAG [4] and GraphRAG [5] model entity relationships but disregard document hierarchy; RAPTOR [6] builds summary trees via recursive clustering but relies on embedding-only retrieval; PageIndex [7] exploits structure but lacks semantic reasoning; legal ontology systems [8,9] formalize domain knowledge but remain disconnected from retrieval pipelines. To address these limitations, a three-tier knowledge graph RAG is proposed that unifies structural navigation, semantic retrieval, and rank fusion: Tier 1 performs LLM-guided tree traversal, Tier 2 applies dual-level semantic retrieval with intent-aware Personalized PageRank, and Tier 3 merges outputs through Reciprocal Rank Fusion with knowledge graph expansion.

This paper makes three contributions. First, a three-tier RAG architecture is designed that unifies structural tree traversal, semantic graph retrieval, and reciprocal rank fusion into a single pipeline. Second, a Vietnamese legal knowledge graph comprising 5,936 entities and 5,963 relations is constructed through a semi-automatic process combining LLM extraction with expert refinement. Third, a multi-intent Personalized PageRank algorithm is introduced that adapts edge weights to detected query intent, allowing the graph to emphasize different relation types for penalty, procedural, or definitional questions. Experiments on 530 questions across three domains show that the system reaches 92.2% Hit@10 on legal QA, surpassing PageIndex by 7.4% and RAPTOR by 51.2%, and 88.5% on education regulation, exceeding PageIndex by 13.9%.

2. Related Work

RAG and KG-augmented retrieval. RAG [1] grounds generation in retrieved evidence. Since early dense retrieval [10], designs have incorporated reflection [11], corrective retrieval, and hypothetical document generation [12]; a survey [3] traces this progression. GraphRAG [5] and LightRAG [4] introduce knowledge graphs for multi-hop reasoning but treat documents as bags of entities, losing structural context. Personalized PageRank [13,14] computes query-specific node importance. RAPTOR [6] builds summary trees via recursive clustering but discards original document hierarchy and uses embedding-only retrieval. PageIndex [7] uses LLM-guided tree traversal for high precision but misses content outside the traversal path.

Legal NLP. Legal IR has moved from lexical methods [15,16] toward domain-adapted transformers such as LEGAL-BERT [17], which capture legal phrasing but treat documents as flat passages. Legal-Onto [8] combines the Rela-model [18] with conceptual graphs for Vietnamese Land Law but relies on manual inference rules. For Vietnamese, PhoBERT [19] and VnCoreNLP [20] provide foundational NLP; Ngo et al. [21] combined BM25 with BERT [22] and domain heuristics but lack graph-based reasoning. Surveys [23,24] confirm that bridging formal legal knowledge with scalable retrieval remains open.

Positioning. No existing system combines ontological knowledge, graph-based retrieval, and structural tree navigation for Vietnamese legal text. NaiveRAG lacks graph and hierarchy; LightRAG [4] and GraphRAG [5] add knowledge graphs but lose hierarchical context; RAPTOR [6] builds summary trees but discards document structure; PageIndex [7] navigates trees but has no knowledge graph; Legal-Onto [8] formalizes legal knowledge but was not designed for retrieval. The proposed system integrates all three capabilities with intent-aware reasoning and rank fusion.

3. Methodology

3.1. Problem Formulation

Given a legal corpus of m documents, each with a rigid hierarchy (Document $\rightarrow$ Chapter $\rightarrow$ Section $\rightarrow$ Article $\rightarrow$ Clause $\rightarrow$ Point), and a user query q, the task is to retrieve a ranked list of the k most relevant articles. The retrieval must operate at *article-level* granularity, as Vietnamese legal practice requires citing specific articles rather than text passages. This differs from standard passage retrieval in two aspects: (1) the search space is a forest of hierarchical trees rather than a flat document collection, and (2) answering a query may require multi-hop reasoning across articles in different chapters or documents linked by cross-references.

3.2. System Overview

Figure 1 illustrates the architecture, which separates into an offline phase (left) and an online phase (right). The offline phase parses legal documents into hierarchical trees, extracts entities and relations into a knowledge graph, and generates summaries for each structural level. The online phase processes each query through four stages: query analysis (Stage 0), LLM-guided tree traversal (Tier 1), dual-level semantic retrieval over the knowledge graph (Tier 2), and fusion with knowledge graph expansion (Tier 3).

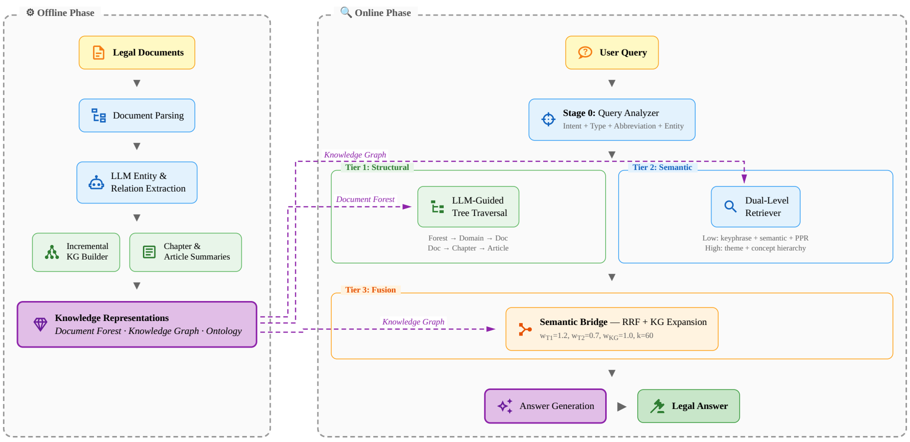

Figure 1. Architecture of the proposed three-tier retrieval system. Offline phase (left): knowledge model construction from legal documents. Online phase (right): query processing through Tier 1 (structural tree traversal), Tier 2 (dual-level semantic retrieval), and Tier 3 (RRF fusion with KG expansion).

3.3. Offline Phase: Knowledge Construction

The offline phase builds three knowledge representations from the raw corpus, following knowledge engineering practice [9].

Document forest. Each legal document is parsed into a rooted tree with six hierarchical levels. Cross-reference edges between documents are extracted, forming a forest that spans multiple legal domains. Documents are organized into domain groups (enterprise law, traffic safety, education regulation) to enable domain-aware routing during retrieval.

Knowledge graph. Entity-relation extraction is performed via one LLM call per article, producing evidence-grounded triples with confidence scores. The resulting graph contains 14 entity types (Action, Legal Term, Organization, Legal Reference, etc.) linked by 56 relation types (*requires*, *applies_to*, *has_authority*, *references*, etc.). Vietnamese slug normalization and 4-pass deduplication ensure consistency across documents. Selected predicates are declared symmetric or transitive to support inference during graph traversal. Construction is semi-automatic: after LLM extraction, evaluation on a development set revealed missing cross-document references; a domain expert then added $\sim$130 relations linking legal terms to governing articles across documents.

Summaries. Chapter-level and article-level summaries are generated for Tier 1 tree traversal, enabling the LLM to select relevant branches without reading full article text.

3.4. Online Phase: Three-Tier Retrieval

3.4.1. Stage 0: Query Analysis

The query analyzer performs: (1) intent classification into six types (Penalty, Definition, Procedure, Requirement, Reference, General) via Vietnamese regex with LLM fallback; (2) abbreviation expansion (e.g., *TNHH* $\rightarrow$ *trach nhiem huu han*); and (3) entity extraction of article references, law references, and legal keywords to seed retrieval. Each query intent amplifies its most relevant relation types during graph traversal (weight 1.0 for primary relations, 0.5 for others).

3.4.2. Tier 1: LLM-Guided Tree Traversal

Tier 1 navigates the document forest using LLM reasoning in three loops. Loop 0 (Forest$\rightarrow$Domain$\rightarrow$Document) matches the query to a domain group using precomputed keywords, then selects the top 5 documents. Loop 1 (Document$\rightarrow$Chapter) selects the top 8 chapters using chapter summaries. Loop 2 (Chapter$\rightarrow$Article) identifies the top 7 articles from article summaries. General Provisions chapters are automatically included. Each loop produces a reasoning trace for interpretability.

3.4.3. Tier 2: Dual-Level Semantic Retrieval

Tier 2 performs hybrid semantic search over the knowledge graph at two levels. The *low level* (entity-specific) combines keyphrase matching, query-entity cosine similarity, and intent-aware Personalized PageRank (PPR). The *high level* (conceptual) identifies legal themes and traverses the ontology class hierarchy, surfacing related articles even without direct entity matches.

The PPR component computes query-specific node importance on the knowledge graph [16]:

$$\mathrm{PPR}(v) = (1-\alpha) \sum_{u \in N(v)} \frac{w(u,v) \cdot \mathrm{PPR}(u)}{d(u)} + \alpha \cdot p(v) \tag{1}$$

where $\alpha = 0.15$ is the teleport probability, $w(u,v)$ is the edge weight adjusted by query intent, $d(u)$ is the weighted out-degree, and $p(v)$ is the personalization vector seeded from matched entities.

3.4.4. Tier 3: Semantic Bridge (RRF Fusion)

Tier 3 merges results from Tier 1 and Tier 2 using Reciprocal Rank Fusion [25]:

$$\mathrm{RRF}(d) = \sum_{s \in \{T1,T2,KG\}} w_s \cdot \frac{1}{k + \mathrm{rank}_s(d)} \tag{2}$$

where $k = 60$ and source weights are $w_{T1} = 1.2$, $w_{T2} = 0.7$, $w_{KG} = 1.0$. Tree traversal receives the highest weight as it provides the most reliable structural signal; the lower weights for Tier 2 and KG allow them to compensate when the LLM selects suboptimal chapters.

After fusion, results are expanded through the knowledge graph's relation edges. Strong relations (*references*, *requires*, *depends_on*) are expanded both cross-chapter (up to 5 articles) and same-chapter (up to 2 articles), with a semantic relevance filter (cosine similarity ≥ 0.4) to suppress noise. Adjacent articles (± 2) from each result are added with decay factor 0.7, as correct answers often lie in neighboring articles. The final ranked list is passed to an LLM for answer generation.

4. Implementation

The system is implemented in Python 3.11 with the following technical choices.

Document parsing. Custom regex parsers detect Vietnamese legal hierarchy markers (“Chuong”, “Muc”, “Dieu”, “Khoan”, “Diem”) and build the document forest. Cross-references between documents are extracted via pattern matching on article citations.

Summary generation. Chapter-level and article-level summaries are generated via LLM (Claude 3.5 Haiku), producing keyword-rich descriptions that Tier 1 tree traversal uses for branch selection without reading full article text.

Knowledge graph construction. Entity extraction and relation identification use Claude 3.5 Haiku[2] via the Anthropic API, with one LLM call per article. Vietnamese slug normalization handles tonal variations (e.g., “cong ty” with and without diacritics). Four deduplication passes merge equivalent entities across documents. The domain ontology defines 14 entity types using OWL classes with a 4-level hierarchy and Vietnamese label mappings.

Online query processing. Tree traversal and query analysis use Claude Sonnet 4 via the Anthropic API. Sentence embeddings are computed with paraphrase-multilingual-MiniLM-L12-v2[3], a 384-dimensional multilingual model from the Sentence-Transformers

[2] `https://www.anthropic.com/claude`

[3] `https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`

library [2]. Vector similarity search uses FAISS[4] for efficient dense retrieval. Personalized PageRank is computed iteratively (max 50 iterations, tolerance 10^{-6}).

Caching and reproducibility. All LLM responses are cached in SQLite, ensuring identical results across runs and eliminating redundant API calls during evaluation.

5. Experiments

5.1. Experimental Setup

The evaluation corpus consists of 18 Vietnamese legal and regulatory documents spanning three domains. The enterprise law domain includes Law 59/2020/QH14 (Enterprise Law) and nine associated decrees, totaling 77 chapters and 840 articles. The traffic safety law domain covers Law 36/2024/QH15 and two penalty decrees. The education regulation domain comprises five postgraduate documents from UIT and VNUHCM, containing 109 articles. In total, the corpus contains 949 articles.

For evaluation, 530 questions were collected from LuatVietnam, ThuVienPhapLuat, and university portals, with expert annotations identifying 1–8 relevant articles per question (average 2.3). The questions are distributed across enterprise law (197), traffic safety law (203), and education regulation (130).

Table 1 summarizes the constructed knowledge graph.

Table 1. Knowledge graph statistics.

Entity Type	**Count**	**Relation Type**	**Count**	**Structure**	**Count**
Total Entities	5,936	Total Relations	5,963	Tree Nodes	9,226
Action	1,903	requires	1,377	Documents	13
Legal Term	1,632	applies_to	810	Chapters	77
Organization	500	has_authority	506	Articles	840
Legal Reference	451	references	434	Cross-refs	168
Other (9 types)	1,450	Other (52 types)	2,506		

Metrics. Since each query has expert-annotated relevant articles, we adopt standard retrieval metrics [16] that enable objective, reproducible evaluation against known ground truth, rather than LLM-as-judge scores used in open-ended generation benchmarks. Hit@$k = \frac{1}{|Q|}\sum_q \mathbf{1}[\exists a \in R_q : \mathrm{rank}(a) \leq k]$ measures the fraction of queries for which at least one relevant article appears in the top-k results, where R_q denotes the set of relevant articles for query q. Mean Reciprocal Rank is defined as MRR $= \frac{1}{|Q|}\sum_q (\min_{a \in R_q} \mathrm{rank}(a))^{-1}$, averaging the reciprocal rank of the first relevant result. Recall@k, Precision@k, and NDCG@k [16] are also computed. Results are reported at $k \in \{1,3,5,10,20,30\}$, with Hit@10 and MRR as the primary metrics. Hit@10 is prioritized because in a RAG pipeline the LLM provides an additional validation layer over retrieved context, at least one correct article among ten candidates suffices for accurate answer generation while limiting noise that could induce hallucination. MRR complements Hit@10 by rewarding higher placement of relevant articles, reducing irrelevant context that the LLM must filter.

[4] `https://github.com/facebookresearch/faiss`

5.2. Baselines

BM25 [15] ($k_1 = 1.2$, $b = 0.75$); TF-IDF [16] with syllable-level tokenization and Vietnamese stopword removal; NaiveRAG [2] (dense retrieval with multilingual sentence-transformers, no reranking); LightRAG[5] [4] with Vietnamese entity extraction and default parameters, using the same LLM backend as the proposed system; RAPTOR[6] [6] with collapsed-tree retrieval using LLM-generated summaries (Sonnet 4 for summarization, MiniLM for embeddings); PageIndex[7] [7] with two-step hierarchical tree search (document→chapter selection, then chapter→article selection), using heuristic summaries extracted from article text. All baselines use Sonnet 4 for online queries and retrieve from the full corpus with max_results=30.

5.3. Main Results

The proposed system achieves 92.2% Hit@10 (MRR 0.603), outperforming PageIndex by 7.4% and the best traditional baseline (Keyword) by 43.7% (Table 2). PageIndex (84.8%) benefits from tree-based structural navigation but misses cross-reference articles outside the traversal path. Notably, RAG baselines (NaiveRAG 32.8%, LightRAG 39.8%, RAPTOR 41.0%) underperform even BM25 (47.5%). This is consistent with findings on domain-specific benchmarks [16]: general-purpose embeddings lack Vietnamese legal vocabulary, and generic KG extraction introduces noise rather than precision. RAPTOR's summaries further lose article-level granularity. BM25 succeeds through exact term matching but lacks structural awareness. The proposed system overcomes both limitations via LLM-guided navigation and domain-specific knowledge graph.

Table 2. Legal domain Hit@k (%) on 400 questions. Best in bold, second underlined.

Method	Hit@5	Hit@10	Hit@20	Hit@30	MRR
Traditional retrieval					
BM25	30.8	47.5	66.2	74.8	0.200
TF-IDF	29.2	45.8	64.5	74.8	0.207
Keyword	33.5	48.5	63.2	72.8	0.222
RAG systems					
NaiveRAG [2]	22.8	32.8	51.8	60.5	0.146
LightRAG [4]	25.5	39.8	47.8	49.2	0.173
RAPTOR [6]	25.2	41.0	54.5	62.3	0.170
PageIndex [7]	<u>76.2</u>	<u>84.8</u>	<u>90.2</u>	<u>91.0</u>	<u>0.559</u>
3-Tier (ours)	**74.5**	**92.2**	**94.8**	**95.0**	**0.603**

5.4. Multi-Domain Performance

Table 3 reports per-domain results, including education as a generalization test.

[5] `https://github.com/HKUDS/LightRAG`
[6] `https://github.com/parthsarthi03/raptor`
[7] `https://github.com/pdfToTree/PageIndex`

Table 3. Per-domain Hit@10 (%) across three domains.

Method	**Enterprise (197)**	**Traffic (203)**	**Education (130)**
NaiveRAG	29.9	35.5	50.0
LightRAG	20.3	58.6	53.1
RAPTOR	18.5	63.5	53.8
BM25	31.5	63.1	70.0
TF-IDF	39.1	52.2	68.5
Keyword	20.3	75.9	56.2
PageIndex	76.6	92.6	74.6
3-Tier (ours)	**86.5**	**98.0**	**88.5**
Ours vs. PageIndex	*+9.9*	*+5.4*	*+13.9*

The system outperforms all baselines across all three domains. Traffic law achieves the highest Hit@10 (98.0%) due to direct violation-to-penalty mappings. Enterprise law is harder (86.5%) due to cross-decree references. Education regulation (88.5%) demonstrates generalization: the largest margin over PageIndex (+13.9%) occurs here, where semantic retrieval captures terminology that tree navigation misses. No domain-specific tuning was applied.

5.5. Analysis

On the legal domain, Hit@10 ranges from 0% (household business, 4 queries referencing articles outside the corpus) to 100% (speed-related penalties), with most categories above 85%. Company establishment (68.4%) remains challenging due to cross-decree dependencies.

A multi-hop example illustrates tier cooperation: "If a member borrows money to meet charter capital and later withdraws it, what are the penalties?" requires linking concepts across chapters. Tier 1 reaches Art. 47 (member obligations) via tree traversal; PPR walks *requires* edges from "charter capital" to Art. 16 Cl. 5 (prohibited acts) in a different chapter; RRF places both in the top 5. No baseline answers this query at Hit@30.

Of the 31 missed queries (7.8%), 27 involve enterprise law. Primary failure modes: company establishment across multiple decrees (6 queries), household business outside the corpus (4 queries), and multi-article queries (>2 expected) accounting for 14 of 31 misses.

6. Conclusion

This paper presented a three-tier knowledge graph RAG architecture for Vietnamese legal question answering. The central contribution is a unified retrieval pipeline combining structural tree traversal, semantic graph search with intent-aware Personalized PageRank, and Reciprocal Rank Fusion over a domain-specific knowledge graph. Experiments across three domains demonstrate consistent superiority over all baselines without domain-specific tuning.

Limitations include Tier 1 latency (approximately 7 seconds per query due to multiple LLM calls); however, in the legal domain, retrieval accuracy takes priority over response speed. A remaining limitation is lower performance on enterprise law queries requiring multi-document cross-references. Future work includes extension to additional regulatory domains, multi-turn conversational support, and explainable retrieval traces.

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